

UNIVERSITY OF CALIFORNIA, DAVIS
Graduate School of Management

Big Data — BAX-423 | Spring 2026

INSTRUCTOR

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Office Hours: Every Sunday, 4:00 – 5:00 PM

Feel free to reach out via email or request an additional meeting.

TEACHING ASSISTANTS

Chris de Mena

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Office Hours: Tuesdays, 2:00 PM – 3:00 PM

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Pavan Kumar Nuthi

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CLASS MEETINGS

Dates: Mar 28; Apr 2*; Apr 11; Apr 18; Apr 25; May 2; May 9; May 16; May 23; May 30 (Wrap-up & Final Exam Discussion); Jun 6 (Final Exam Submission)

Times: Section 1: 9:00 AM – 12:30 PM | Section 2: 1:30 PM – 5:00 PM

Location: 123

Finals: Jun 6 — 1-week AI-assisted take-home final exam.

* **Note:** The second lecture (Apr 2) will be held on Thursday.

LAB SECTION

Lab sessions are recorded and shared. Attendance is not mandatory. Occurs every Tuesday (starting 31st March) from 1 - 2 PM, [Join Lab Zoom](#)

Course Overview

This course focuses on the **design, implementation, and optimization of large-scale machine learning systems** used in production environments such as search engines, recommendation platforms, advertising systems, and real-time data pipelines.

Students will learn to process large datasets using **PySpark**, design streaming systems with **Kafka**, and build modern **recommendation, ranking, and reinforcement learning systems**. The course also introduces **Large Language Models (LLMs)** from a systems perspective.

Emphasis is placed on performance, scalability, system tradeoffs, and real-world deployment constraints.

Learning Objectives

By the end of this course, students will be able to:

- Build scalable batch and parallel pipelines using PySpark
- Apply hashing, sampling, and sketching techniques for large-scale data
- Design streaming data systems using Kafka

- Design embedding-based retrieval and recommendation systems
- Understand LLMs as large-scale data pipeline consumers
- Apply reinforcement learning methods in large-scale decision systems

Prerequisites

1. Prior exposure to probability theory
2. Prior coding experience with Python
3. Knowledge of basic computer programming concepts

Course Schedule

The schedule below is tentative and subject to change. All changes will be announced on Canvas.

Date	Topic & Content	Due
Mar 28	Modern ML Development & AI-Assisted Systems – ML systems: data → systems → models → decisions – Using AI tools (Cursor) for pipelines & refactoring – Prompting vs fine-tuning (conceptual) – ML system design	
Apr 2*	Data Sketching & Efficient Learning – Hashing & consistent hashing – Sampling (reservoir, mosaic) – Approximate data structures (Bloom filters, Count-Min Sketch, HyperLogLog)	
Apr 11	Kafka & Streaming Data Systems – Kafka architecture, topics, partitions, offsets – Delivery semantics (at-least-once vs exactly-once) – Streaming features for ML systems	HW 1 Due
Apr 18	Scalable Data Processing with PySpark – Spark architecture & execution model – DataFrames and distributed data operations – Transformations vs actions, DAG execution – Aggregation, joins, and shuffle operations	HW 2 Due
Apr 25	Embeddings & Dimensionality Reduction – Curse of dimensionality – Johnson–Lindenstrauss theorem – Embeddings as compressed representations – Word2Vec & representation learning	HW 3 Due
May 2	Recommendation Systems – Recommender problem framing – Collaborative filtering & latent factor models	HW 4 Due

Date	Topic & Content	Due
	- Two-tower models - Candidate generation at scale	
May 9	Ranking & Multi-Stage Recommendation Systems - Ranking vs recommendation - Multi-stage pipelines (candidate gen → ranking → re-ranking) - Pointwise / pairwise / listwise ranking - Evaluation: Precision@K, Recall@K, NDCG	HW 5 Due
May 16	Introduction to Reinforcement Learning - Recommendations as sequential decisions - Feedback loops & delayed rewards - Exploration vs exploitation - Q-learning intuition & limits	HW 6 Due
May 23	Deep Reinforcement Learning & DQN - From Q-learning to function approximation - Deep Q-Networks (DQN) & experience replay - Applications to feeds, ranking, and engagement systems	HW 7 Due
May 30	Course Wrap-Up & Final Exam Discussion - Course overview & key takeaways - Ethics in large-scale ML systems - Career pathways & advice - Final exam Q&A and discussion	HW 8 Due
Jun 6	Final Exam Submission 1-week AI-assisted take-home exam due	

Lab Schedule

Lab sessions are held weekly and recorded for asynchronous access. Attendance is not mandatory. [Join Lab Zoom Session](#)

Week	Lab Topic
Week 2	Fine-tuning & AI Agents
Week 3	Hashing & Streaming
Week 4	Kafka
Week 5	PySpark
Week 6	Text Embeddings
Week 7	Recommendation Systems
Week 8	Recommendation Systems II: Rankings
Week 9	Deep Reinforcement Learning

Grading

Component	Weight
Homework Assignments (8 total)	64%
Final Examination (1-week take-home)	28%
In-Class Quizzes (8 total)	8%

Course Materials

Textbooks

This course does not follow a single required textbook. The following titles are recommended as optional reference material. None are required for assignments or the final exam.

4. Designing Data-Intensive Applications — Martin Kleppmann (optional)
5. Mining of Massive Datasets — Leskovec, Rajaraman & Ullman (optional — freely available online at mmds.org)
6. Reinforcement Learning: An Introduction — Sutton & Barto (optional — freely available online)
7. Speech and Language Processing — Jurafsky & Martin, 3rd ed. (optional — Chapter 6 covers embeddings; available online)

Textpak

No textpak or course reader is used in this course. All readings and lecture materials will be posted directly to Canvas.

Lecture Topics & Notes

Lecture slides will be posted on Canvas before each class session.

Lecture 1 — Modern ML Development & AI-Assisted Systems

- Course overview: data → systems → models → decisions
- Using Cursor for ML systems development (prompting for pipelines, debugging & refactoring)
- Prompting vs fine-tuning (conceptual overview)
- Why most production ML systems avoid fine-tuning
- ML systems design case study

Lecture 2 — Data Sketching & Efficient Learning

- Hashing fundamentals & consistent hashing

- MapReduce mental model (framework-agnostic)
- Streaming constraints: one pass, limited memory
- Sampling: reservoir sampling, mosaic sampling
- Approximate data structures: Bloom filters, Count-Min Sketch, HyperLogLog

In-Class Lab: Implement consistent hashing; reservoir sampling; exact vs approximate counting

Lecture 3 — Kafka & Streaming Data Systems

- Kafka architecture: topics, partitions, offsets
- Delivery semantics: at-least-once vs exactly-once
- Kafka as backbone for ML systems
- Introduction to PySpark (overview)

Lecture 4 — Scalable Data Processing with PySpark

- Spark architecture & execution model: driver, executors, clusters, lazy evaluation
- DataFrames and distributed data operations
- Transformations vs actions; DAG execution
- Aggregation, joins, and shuffle operations

Lecture 5 — Embeddings & Dimensionality Reduction

- Curse of dimensionality
- Johnson–Lindenstrauss theorem & intuition
- Embeddings as compressed representations
- Word2Vec & representation learning

Lecture 6 — Recommendation Systems

- Recommender system problem framing
- Collaborative filtering & latent factor models
- Two-tower models
- Candidate generation at scale

Lecture 7 — Ranking & Multi-Stage Recommendation Systems

- Ranking vs recommendation
- Multi-stage pipelines: candidate generation → ranking → re-ranking
- Ranking approaches: pointwise, pairwise, listwise
- Models: Deep & Wide; Deep & Cross
- Evaluation: Precision@K, Recall@K, NDCG

Lecture 8 — Introduction to Reinforcement Learning

- Recommendations as a sequential decision problem
- Feedback loops & delayed rewards
- Exploration vs exploitation

- Markov Decision Process intuition
- Q-learning update rule and why tabular Q-learning breaks at scale

In-Class Lab: Simple recommender as a bandit / Q-learning problem; reward design experiment

Lecture 9 — Deep Reinforcement Learning & DQN

- From Q-learning to function approximation
- Deep Q-Networks (DQN) & experience replay
- Applications: feed ranking, content recommendations, notifications & engagement systems

In-Class Lab: Implement DQN; apply to toy recommendation or ranking environment

Assignments & Policies

Homework Assignments

There are **8 homework assignments**, due approximately one per week starting after Lecture 2. Each assignment includes questions and/or hands-on coding tasks in Python. Assignments are completed in **groups of 2**. Groups must submit their own independent work, though conceptual discussion with other classmates is permitted.

Assignments are posted and submitted on Canvas by 11:59 PM on the due date.

Note: If you have an odd-enrollment situation or your partner drops the course, contact the CA promptly to discuss alternatives.

Late Assignment Policy

- Up to 24 hours late: 25% grade reduction
- 24–48 hours late: 50% grade reduction
- More than 48 hours late: Not accepted

In-Class Quizzes

There will be 8 short in-class quizzes throughout the quarter, held during lecture sessions. Dates will be announced in advance on Canvas. There is no make-up policy for missed quizzes unless a documented medical or family emergency is presented to the instructor.

Final Examination

The final exam is a **1-week take-home assessment** (due Jun 6) designed to reflect real-world data science and machine learning workflows. Students may use AI-assisted tools including code assistants, LLMs, and documentation.

- Students must submit the prompts they used (including key iterations) as part of the final submission
- Submissions must demonstrate original understanding, clear reasoning, and ownership of the solution

- Students must be prepared to explain design choices, assumptions, and results
- Any unacknowledged copying — human- or AI-generated — is an academic integrity violation

Re-grading Policy

If you believe a grading error has occurred, submit a formal memo to the CA describing the error within one week of the assignment being returned. Include supporting documentation (class notes, slides, etc.). If unresolved, escalate to the instructor by email with all relevant material.

Questions & Discussion

The Course Assistant is the first point of contact for homework questions and will respond in a timely manner. Class discussions are held on Piazza.

Academic Integrity

| Notice of the Code of Academic Conduct (UC Davis Regulation 537)

All students enrolled in this course are governed by the UC Davis Code of Academic Conduct, which sets forth responsibilities for students and faculty to maintain a spirit of academic honesty and integrity. Please review the Code at:

ossja.ucdavis.edu/code-academic-conduct. Violations may result in disciplinary action. Further details may be obtained from the GSM Associate Dean or the Office of Student Support and Judicial Affairs.

Given the AI-assisted nature of this course, students are especially reminded that **uncritical or unacknowledged submission of AI-generated work constitutes a violation** of the Code.

Statement on Accommodation

Students requiring accommodation for a disability should contact the [Student Disability Center](#) at sdcc@ucdavis.edu or 530-752-3184. Once you receive your Letter of Accommodation, please submit it to the instructor as soon as possible and within the first two weeks of the course.